

# Jianlun (Lody) Sun, MSc.

✉ jianlun.sun@helsinki.fi

in linkedin.com/in/lodysun/

🌐 <https://lodysun.github.io>

## Research Interest

I am interested in how the brain constructs and maintains meaning over time. Experience arrives continuously, yet we perceive it as structured: segmented into events, linked by causal and temporal relations, and organised into wholes that can be recalled, revised and told to others. My work asks how these structures are built during ongoing experience, how they are represented in the brain, and how they are transformed as perception becomes memory. I approach this with naturalistic paradigms and computational methods, combining behavioural modelling, EEG, fMRI and language-model analysis, and I take up new methods when a question requires them.

## Education

Sep 2024 – Jul 2026

📖 **M.Sc. Neuroscience, University of Helsinki** (GPA: 4.35/5)

Thesis title: *Staged Rule Acquisition in WishGLD: Dissociation and Compression Across Learning*. (Grade: 4/5)

- Independently developed a pair-resolved fMRI analysis framework for Sherlock, integrating viewing and free-recall representations, constrained permutation testing, and time-resolved RSA to show selective reinstatement of earlier narrative context in posterior medial cortex during recall.
- Developed a trial-resolved analysis framework for a naturalistic set-shifting task (WishGLD), combining behavioral modeling, two-state HMM trajectory markers, and feedback-locked EEG to characterize how rule acquisition unfolds from search to stabilized performance.
- Identified that rule acquisition is not a unitary event: logical hypothesis collapse, first correct response related neural confirmation, and later behavioral stabilization can dissociate, with early learning showing a clearer staged transition structure that becomes compressed with task exposure.

Sep 2018 – Jun 2022

📖 **B.Sc. Biological Sciences, ShanghaiTech University**

Thesis title: *Exploring the relationship between Locus Coeruleus integrity and set-shifting dysfunction in Parkinson's Disease*.

## Research Publications

### Conference & Symposium Presentations

- 1 J. Sun, "When payoff returns: Reinstatement of earlier narrative context during recall," in *Brain & Mind Symposium 2026*, Poster presentation, Helsinki, Finland, Oct. 2026.
- 2 J. Sun, K. Lukander, and B. U. Cowley, "Staged rule acquisition in WishGLD: Dissociation and compression across learning," in *9th International Conference on Motivational and Cognitive Control (MCC)*, Poster presentation, Helsinki, Finland, Aug. 2026.

# Research Experience

- Jun 2025 – Oct 2026

**Research Assistant**, HiPerCog Group, Department of Education, University of Helsinki (Supervisor: Ben Cowley).
- Designed and implemented a modified WCST in PsyToolkit with an orthogonal orientation rule, enabling future reinforcement-learning modeling of multidimensional strategy use.
  - Built the human-side analysis pipeline for a rat-human Go/NoGo attentional set-shifting paradigm: preprocessing from raw behavioral output, and a model-fitting pipeline matched to the existing rodent pipeline so that both species could be fitted under a single model.
- Mar 2021 – Jun 2022

**Undergraduate Research Assistant**, Institute of Neuroscience, Chinese Academy of Sciences (Supervisor: Ye Zheng)
- Quantified Locus Coeruleus/Substantia Nigra integrity from neuromelanin-MRI via manual ROI tracing and intensity normalization
  - Linked LC structural metrics to set-shifting deficits in Parkinson's disease (CANTAB IED)
  - Assisted in collecting and organizing neuropsychological data from DBS patients, contributing to analyses of stimulation-related cognitive impairment in GPi/STN cohorts. (Collaborator: Dr. Wu Jianjun)

# Skills

- Languages

Mandarin (Native), English [TOEFL iBT: 112/120 (R28, L30, W30, S24)]
- Coding

Python, R, MATLAB
- Analytical Skills

Python (pandas, NumPy, scipy, matplotlib);  
EEG (MNE-Python, EEGLAB, ERP/time-frequency analysis, ISC, IS-RSA);  
Statistics/modeling (GLM, PCA, HMM, mixed-effects models, permutation testing, FDR correction);  
fMRI (GLM, MVPA/decoding, RSA - independently implemented)  
NLP (Basic)

# Academic Awards

- 2024-2026

**100% Tuition Fee Waiver**, University of Helsinki, Master's Programme in Neuroscience

## Research-related Activities

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Oct 2025 - Jun 2026

📖 **Curriculum Reviewer**, Master's Programme in Neuroscience, University of Helsinki

- Selected student reviewer for evaluating and redesigning the online course **Systems & Cognitive Neuroscience**.
- Provided pedagogical feedback on structure, clarity, and accessibility for students with diverse academic backgrounds.
- Contributed to improving learning alignment and conceptual scaffolding for future cohorts.

Jan 2025 - Jun 2026

📖 **Student Representative**, Master's Programme in Neuroscience, University of Helsinki

- Served as student representative on the programme board; communicated student needs and coordinated curriculum improvements.
- Synthesized student feedback into actionable recommendations that informed course design, assessment methods, and program structure.